

ORIGINAL RESEARCH

Markov-GAN: Markov image enhancement method for malicious encrypted traffic classification

Zhangguo Tang^{1,2}  | Junfeng Wang³ | Baoguo Yuan³ | Huanzhou Li² | Jian Zhang² | Han Wang²

¹School of Cyber Science and Engineering, Sichuan University, Chengdu, China

²School of Physics and Electronic Engineering, Sichuan Normal University, Chengdu, China

³College of Computer Science, Sichuan University, Chengdu, China

Correspondence

Junfeng Wang, College of Computer Science, Sichuan University, Chengdu 610065, China.
Email: wangjf@scu.edu.cn

Funding information

Basic Research Program of China, Grant/Award Number: 2019-JCJQ-ZD-113; National Key Research and Development Program, Grant/Award Numbers: 2018YFB0804503, 2019QY1400; National Natural Science Foundation of China, Grant/Award Numbers: U1836103, U20A20161

Abstract

The rapidly growing encrypted traffic hides a large number of malicious behaviours. The difficulty of collecting and labelling encrypted traffic makes the class distribution of dataset seriously imbalanced, which leads to the poor generalisation ability of the classification model. To solve this problem, a new representation learning method in encrypted traffic and its diversity enhancement model are proposed, which uses the diversity of images to represent the diversity of traffic samples. First, the encrypted traffic is transformed into Markov images. Then, a diversity maximisation Markov-GAN based on the Simpson index is designed to generate new Markov images. Finally, the balanced Markov image set is sent to the CNN for classification. Experimental results show that the proposed method can predict the whole dataset space with only a few original samples. And the classification accuracies under different imbalance degrees are significantly improved, all of which are over 90%. The enhanced Markov image set can effectively alleviate performance generalisation deviation caused by different network depths. Even an ordinary CNN has almost the same classification effect as VGG13 and VGG16. Compared with other data enhancement methods, the Markov-GAN only needs to balance the transform domain dataset, which is lightweight, easy to train and has stronger amplification ability.

KEYWORDS

class imbalance, generalisation ability, malicious encrypted traffic, Markov image, Markov-GAN, representation learning

1 | INTRODUCTION

With the promotion of global industrial digital transformation and the development of mobile network and IoT, a large number of network services and applications use encryption technology as a priority method to ensure information security [1]. Gartner predicted in 2019 that 80% of website traffic will be encrypted [2]. Traffic encryption not only protects privacy but also allows malicious behaviour to escape the supervision of traditional security detection technology. More than 70% of malware activities use some type of encryption to hide malware transmission, control command activity and data leakage [3]. New Zscaler threat research reveals the emerging techniques

and impacted industries behind a 260-percent spike in attacks using encrypted channels to bypass legacy security controls [4]. Cyren [5] also found that the attacks of ransomware using encrypted traffic have increased by 5 times, and each major ransomware family has spread through HTTPS. In the field of mobile applications, malicious APPs generally use encrypted traffic (such as HTTPS) to transmit network data to avoid detection. More than 30% of SSL-based attacks deceived trusted cloud providers such as Dropbox, Google, Microsoft and Amazon to distribute malware through encrypted channels, which has become more and more complex in avoiding detection [6]. Therefore, how to effectively identify malicious traffic has become an important challenge to network security.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2022 The Authors. *IET Information Security* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

There are two difficult problems in the identification and classification of encrypted traffic. On the one hand, the characteristics of encrypted traffic have changed or hidden, which makes the traditional traffic detection technology difficult to extract and represent effectively. In order to solve this problem, deep learning methods have been continuously introduced, such as RNN [7], GAN [8], CNN [9], KNN [10] and LSTM [11]. These methods focus on continuously improving the representation learning of malicious features in encrypted traffic: either construct the feature engineering of encrypted traffic, such as spatio-temporal features, head features, load features, statistical features etc. [1, 12] or transform the traffic with different fineness into transform domain forms such as image [13], matrix [14] or N-gram [15]. However, the effects of these encrypted traffic representation learning methods are limited to specific datasets, and there are problems with preprocessing distortion and difficult migration.

On the other hand, the existing malicious encrypted traffic datasets have the class imbalance problem of ‘fat-tail distribution’, which leads to insufficient training and seriously affects the accuracy and generalisation effect of classification. For the datasets applied in some typical papers, the number of samples varies greatly between different categories. For example, in the dataset used by Wang [16], the maximum training sample is 60,000, while the minimum is only 298. H. F. Alan [17] found that the recognition accuracy of the models trained by imbalanced datasets decreased by more than 21% when applied. In order to solve this problem, semi-supervised method [18, 19], unsupervised method [13] and generative method [20] are mainly proposed. Among them, semi-supervised uses a large number of easily available unlabelled encrypted traffic and a small number of artificially labelled encrypted traffic to form a dataset. The generation method generally uses the GAN model [8] to fill the weak samples in the encrypted traffic dataset, and the model learnt on the filled dataset has a higher accuracy. However, these methods not only need labelled data but also consider the matching problem between the original dataset and the labelled dataset. Different datasets have different features, which leads to the reduction of model accuracy and generalisation ability.

In summary, there is neither a unified representation method for encrypted traffic nor a sufficient and balanced public label dataset [1, 12]. In order to solve the normalised representation of encrypted traffic and the generalisation of classification under the condition of imbalanced class distribution, a malicious encrypted traffic classification method based on Markov image feature transformation and enhancement is proposed. The input is the original encrypted traffic session, which is transformed into a normalised Markov image. Then, through Markov-GAN based on maximising population diversity, new and more Markov images are generated. The enhanced and balanced image set is sent to the CNN for recognition and multi-classification. Compared with the traditional encrypted traffic representation methods [10, 21, 22] and data enhancement methods [8, 19, 20, 23], our method has higher classification accuracy, stronger amplification ability and

better generalisation effect. The main contributions of this paper are summarised as follows.

- An encrypted traffic representation method based on a Markov image is proposed, which is the first time in the traffic field. The high-dimensional spatio-temporal features of the original encrypted traffic are represented as Markov images with low-rank structure, which realises the unified representation learning of encrypted traffic datasets with different specifications. This method does not need to slice and fill the encrypted traffic artificially and reduces the preprocessing distortion. Compared with the traditional grey image, the Markov image is lighter and more friendly to the classification model.
- A Markov-GAN based on the compression coding loss function and Simpson exponent is designed. The Markov-GAN not only enhances the texture quality and diversity of Markov images but also uses as few original traffic samples as possible to adaptively expand and balance the image dataset, so as to realise the prediction of the sample space of the whole original dataset.

Experiments have been conducted on multiple encrypted traffic datasets and various imbalance degrees of datasets. In terms of classification accuracy, the proposed method has better effect. The accuracy of multiple datasets exceeds 99%, which is higher than that of similar work. In terms of data enhancement, the dataset is artificially set to a variety of imbalance degrees, and the classification accuracy after enhancement is significantly improved, all of which are over 90%. In terms of generalisation ability, the accuracy of the proposed method on three different datasets is more than 97%, and the accuracy difference between models such as ordinary CNN, VGG13 and VGG16 is only 0.15%. In addition, in terms of performance, compared with grey image, the Markov image does not need segmentation and filling. It has richer texture and lighter weight. The maximum decrease in image size of the same length flow is 98.1%. Experimental results show that the proposed method not only obtains higher classification accuracy but also greatly improves the generalisation ability under imbalanced conditions.

The remainder of this paper is structured as follows: Section 2 reviews the related work of malicious encrypted traffic classification. Then Section 3 details the method of malicious encrypted traffic classification based on the Markov image enhanced by the Markov-GAN and CNN. Experimental evaluation is given in Section 4. Finally, Section 5 concludes this paper.

2 | RELATED WORK

The essence of malicious encryption traffic classification is to learn the feature distribution of data. The stronger the ability of feature compression (dimensionality reduction), the better the accuracy and generalisation effect of classification [24]. Under the condition of imbalanced class distribution, there are

two main dimensions to improve the generalisation ability of deep learning, namely, the representation learning of encrypted traffic and the data enhancement of samples.

2.1 | Representation learning based on feature engineering and transform domain method

2.1.1 | Method based on feature engineering

The method based on feature engineering is to construct useful data features through prior knowledge. The feature engineering of encrypted traffic mainly includes space-time features [10, 25], head features [26], load features [26] and statistical features [27]. Ivan et al. [11] constructed the feature engineering of encrypted traffic, extracted 40 data features of 4 categories, and obtained 94.87% accuracy by using the LSTM classification method. Wang [28] gave six statistical features of TLS (upload byte, download byte size etc.) and four statistical features of HTTPS stream (user agent, request URL etc.). Gil et al. [10] proposed a set of time-related VPN traffic features, such as packet length, transmission time interval, direction etc. Hu et al. [29] constructed the traffic features of the whole session according to the length, arrival time interval and packet text distribution of the first 150 data packets of the TLS session. Anderson et al. [30] extracted the features of 18 types of malicious encrypted traffic from the text information of encryption protocol packets, including encryption suite, compression function, supported version and other parameters and achieved an accuracy of 90.3%. Nychis et al. [31] studied the distribution of encrypted traffic header features and behaviour features based on entropy and pointed out that it is more effective to use time series to construct feature engineering.

The feature engineering of encrypted traffic directly affects the classification accuracy and generalisation effect [32]. Although it has achieved high accuracy in a specific experimental dataset, this method heavily relies on artificial prior knowledge and the dimension of feature extraction is limited, which leads to over-fitting of classification model and poor generalisation ability.

2.1.2 | Transform domain method based on feature self-learning

In order to solve the problem of difficult feature extraction of encrypted traffic data and enhance the ability of representation learning and generalisation, a transform domain method based on feature self-learning is proposed, that is, the neural network automatically learns the useful features hidden in the data.

Feature self-learning based on traffic image transformation: Wang [21] first proposed an encrypted traffic classification method based on traffic image conversion. The first 784 bytes of traffic are converted into grey image, and the image data features are learnt by a two-dimensional CNN. The accuracy is 99.17% on the USTC-TFC2016 dataset. Guo

[13] used the first 1521 bytes of the traffic session as the feature vector and converted it into a 39×39 grey image. The average accuracy of 92.92% is obtained by using a two-dimensional convolutional neural network. Bazuhair et al. [26] used Perlin noise to encode the given connection features into images and trained a deep learning model of connection stream for two classification. The accuracy rate on the CTU-13 dataset has reached 97%.

Feature self-learning based on traffic conversion to Markov chain: Korczynski [22] first proposed an encrypted traffic classification method based on a Markov chain. The SSL/TLS transmission message sequence is abstracted as a time-varying Markov random process. The SSL/TLS message is modelled by the Markov chain, and the true positive rate is 97%. This work has been improved by other researchers [14, 33]. Considering the two-way communication process of encrypted session, an encrypted traffic classification method based on the perceptible attribute of a second-order Markov chain is proposed, which further improves the detection effect.

Feature self-learning based on transforming traffic into vector: Cheng et al. [9] extracted the first 1000 bytes of the traffic session payload as the feature vector and used skip-gram to embed words with a dimension of 300 in each byte of the feature vector to form a new feature vector. The accuracy of 91.03% is obtained by using a one-dimensional convolutional neural network. Wang et al. [15] took the text content of the header of the data packet as a fixed length sentence vector, in which the byte information is regarded as a word, and used the long-term and short-term memory neural network model to learn the temporal relationship between bytes, with an accuracy of 97.2% on the Mirai-RGU dataset. In addition, Wang et al. [35] proposed a traffic feature vector synthesis method based on a stack automatic encoder, which can classify and identify the applications of encrypted traffic.

The transform domain method based on feature self-learning does not need to extract data features manually and has good portability. But the shortcomings are also obvious. For example, it is necessary to slice, intercept and fill traffic data. Most of these processes are based on experience and have natural information loss. In addition, it is difficult to normalise the traffic to grey image. The size of the picture depends heavily on the length of the traffic, which affects the generalisation performance and training efficiency. The Markov chain is difficult to directly input the learning model, and it is difficult to train.

2.2 | Dataset balancing method based on data enhancement

This method uses the existing samples to generate more samples and expands the weak samples to eliminate the imbalance of the dataset, so as to improve the generalisation performance of the model. At present, there are mainly semi-supervised [18] and unsupervised [13] methods. Japkowicz et al. [36] pointed out that imbalanced class distribution would hinder the performance of standard classifiers, and then proposed two dataset balancing methods: RUS and ROS. Chawla

[37] proposed a method combining over-sampling and under-sampling of a few classes to obtain better classification results. Rezaei [19] combined a large amount of unlabelled traffic data and a small amount of artificially labelled traffic data to form a training dataset for pretraining and retraining, respectively, and the accuracy was improved to 84.53%. Groisman [38] instantiates a small number of marked encrypted traffic in the source configuration and generalises it to different configurations, which can identify user operations in different unmarked target configurations.

The more novel idea is the generative model. The advantages of the GAN are used for data enhancement, and its generalisation ability is reflected in high fidelity and diversity. Vu et al. [8] used the AC-GAN to fill the weak samples in the dataset, and many algorithms such as SVM, DT and RF have obtained higher recognition accuracy on the dataset filled successively. Wang [20] used the CGAN to generate encrypted traffic to supplement the samples of vulnerable categories, so as to balance the traffic dataset. Compared with the original traffic dataset, the detection rate is further improved by 1.54%. Further work transformed the GAN discriminator into a classifier SGAN [23], supplemented a small number of labelled samples, and formed a new balanced dataset.

Although these data enhancement methods have achieved certain results, compared with generating images, the GAN needs to ensure its activity and aggressiveness to generate encrypted traffic, which is difficult, and its availability cannot be guaranteed. The generation method based on some encrypted traffic is also difficult to be generalised to other encryption protocols, and its universality is poor. More importantly, compared with the huge sample space of the original encrypted traffic, the amplification ability and effect of directly generating traffic are very limited.

Table 1 gives a simple comparison of existing solutions, mainly focussing on two indicators: dataset imbalance and generalisation ability. As shown in Table 1, the method based on feature engineering has large workload and poor portability. The method based on transform domain does not need to extract features manually, but the preprocessing distortion is large. Data enhancement methods can effectively balance datasets, but the current methods enhance the original traffic, which has huge overhead and limited effect. Therefore, this paper proposes a new transform domain method, which uses the Markov image with low-rank structure to perform unified representation learning on malicious encrypted traffic. The Markov-GAN is used to generate images instead of generating original traffic, and the indirect balance of the original dataset is realised by enhancing the diversity of images.

3 | THE PROPOSED METHOD

The general framework of the proposed method is shown in Figure 1. It is divided into three specific stages: Markov image conversion, image enhancement based on the Markov-GAN and encrypted traffic classification training.

Phase 1: The dataset Pcap file stores the original encrypted traffic. Relevant experimental studies show that session is a more robust traffic representation form [13, 21]. On the one hand, converting sessions into images does not destroy the basic information of traffic. On the other hand, the accuracy of session-based classification is generally higher than that of flow-based classification. Therefore, it is necessary to split the original Pcap file by session and divide the encrypted traffic dataset into different numbers and types of Pcap sub-files. The size of the subfile varies with the length of the session. Then, according to the Markov image conversion algorithm, each traffic session is converted into a Markov image.

Phase 2: This step is the key to solve the problem of dataset imbalance. Because the class distribution of the original traffic dataset is imbalanced, the converted 'original Markov image set' inherits the imbalanced characteristic. Therefore, the function of this stage is to use the proposed Markov-GAN to generate more new pictures and enhance the diversity of the 'original Markov image set'. Then, a 'new balanced Markov image set' is obtained for the later depth learning model to train. It is worth emphasising that it is not the original traffic that is generated, enhanced and balanced, but its transformed domain image.

Phase 3. In this stage, a variety of different CNNs are used as classification models to evaluate the performance of the 'new balanced Markov image set', which is used as training set. The 'original Markov image set' is used as a test set to test the classification effect, generation quality and balance ability of the proposed model.

3.1 | Representation learning of malicious encrypted traffic based on Markov images

Different encrypted traffic sessions have different probability of message state transition. This random process has Markov characteristics and can be used to characterise the temporal and spatial characteristics of encrypted traffic [22, 33]. The specific details of Markov transformation process have been discussed in the relevant literature [34]. For convenience, a visual Markov process of a four byte session sequence is given here. The

TABLE 1 Comparison of classification methods under class imbalance

Method	Dataset	Features	Classification techniques	Weakness
Ref. [10, 11, 28–30]	Imbalance	Feature engineering	SVM, DT, RF	Difficult and poor generalisation ability
Ref. [9, 14, 15, 21]	Imbalance	Transform domain	CNN, LSTM	Relying on experience and large distortion
Ref. [18, 19, 35–37]	Artificial balance	Feature engineering	Semi-supervised and unsupervised method, ROS, Smote	Time consuming and poor portability
Ref. [8, 20, 23]	Machine generated balance	Traffic	GAN + CNN	Complex and poor universality

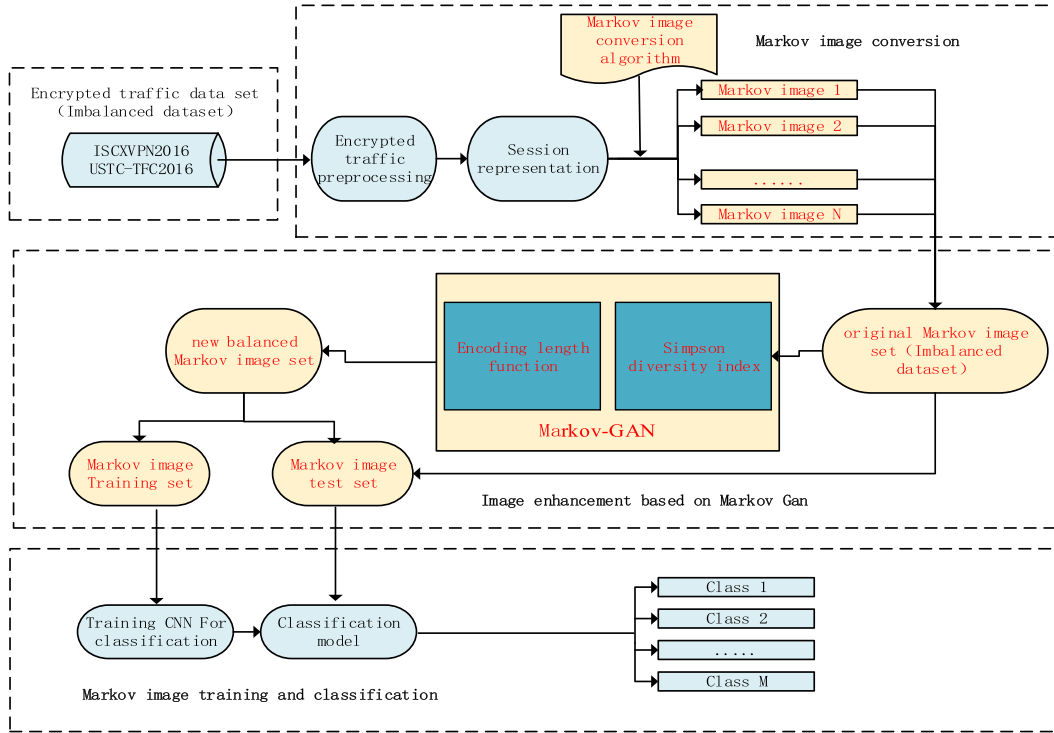


FIGURE 1 Markov image enhancement method for malicious encrypted traffic classification

session is taken from the ISCXVPN2016 dataset, as shown in Figure 2.

The conversation in the figure forms a transition probability sequence from beginning to end. It is cumbersome to calculate the Markov chain matrix directly, and it is difficult to input it directly into the classification model. Therefore, a representation method based on the Markov image is proposed. The state transition of encrypted traffic message is indirectly represented by the conversion between each byte in the corresponding Pcap file, and the transition probability matrix is encoded into the corresponding Markov image. This method has the characteristics of low dimension and scale invariance, that is, the session of traffic with variable lengths can be transformed into a fixed length data format through a Markov image structure. In addition, because this method does not need traffic slicing and filling, the information loss caused by feature extraction is minimised.

The specific method is to first split the original stream into multiple independent Pcap files according to the session. Then, the traffic file is read in binary mode and converted into byte values (0–255) with one byte as a unit. The random process of byte stream converted from encrypted traffic can be expressed as:

$$B_i, i \in \{0, 1, \dots, N-1\} \quad (1)$$

Where, N is the byte length of encrypted traffic, and B_i is the value of the i th byte, $B_i \in \{0, 1, \dots, 255\}$.

Then, calculate the transition probability of two adjacent bytes X_1 and X_2 , that is, the probability $P(X_2|X_1)$ that X_2 appears after X_1 :

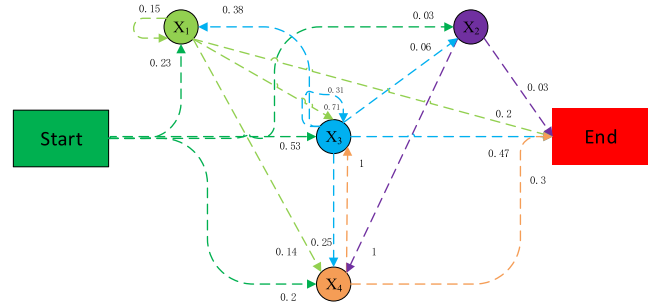


FIGURE 2 A visual Markov process of a four byte session sequence

$$P(X_2|X_1) = \begin{bmatrix} p_{0,0} & p_{0,1} & \cdots & p_{0,255} \\ p_{1,0} & p_{1,1} & \cdots & p_{1,255} \\ \vdots & \vdots & p_{x_1,x_2} & \vdots \\ p_{255,0} & p_{255,1} & \cdots & p_{255,255} \end{bmatrix} \quad (2)$$

Where P_{x_1,x_2} are the transition probabilities, and x_1 and x_2 are the actual values of X_1 and X_2 , respectively, that is $x_1, x_2 \in \{0, 1, \dots, 255\}$. The calculation formula of P_{x_1,x_2} is as follows:

$$P_{x_1,x_2} = \frac{P(x_1, x_2)}{P(x_1)} = \frac{P(X_2|X_1)}{\sum_{j=0}^{255} P(X_j|X_1)} \quad (3)$$

After obtaining the transition probability matrix, take every two adjacent byte values as the coordinate points in the 256×256 size image, and its transition probability value is the value

of the pixel point; then the matrix can be image encoded to obtain the Markov image. The Markov image is based on the ASCII code corresponding to each byte in the Pcap file, and its range is 0–255. Therefore, the maximum number of transition probabilities can be obtained is 65,536. Therefore, the size of Markov graph is fixed, which is a 256×256 transition probability matrix obtained from formula (2).

Because the sessions of the same class of traffic have similar transition probability, the Markov images of the same class are homologous in probability. The process of converting traffic into the Markov image is shown in algorithm 1 and Figure 3, respectively:

Algorithm 1 Convert encrypted traffic into Markov image

• **Input:**

Network traffic (Pcap);

• **Output:**

Markov image (Mi);

```

1: For Pcap in  $\{1, 2, \dots, n\}$  do
2:   Get network traffic quintuple content
3:   Calculate transition probability
   between contents
4:   Estimate transfer probability  $P_{x_1, x_2}$ 
5:   Generate transition probability matrix
 $P(X_2 | X_1)$ 
6:   Image coding of transition probability
   matrix  $\rightarrow$  Mi
7: End For
8: Return Mi

```

3.2 | Generation model based on Markov-GAN

3.2.1 | Architecture and process of the Markov-GAN model

The framework of the Markov-GAN model is shown in Figure 4, including generator G, discriminator D, the DBSCAN clustering network and ‘original Markov image set’. Among them, the DBSCAN clustering network clusters the generated Markov images for the calculation of regularisation terms. The generator and discriminator realise the enhancement and balance of samples by synthesising gradient optimisation and diversity mechanism. The basic process is that after the noise enters the generator G, a series of new Markov image countermeasure samples are generated, and the discriminator D performs ‘true and false discrimination’ based on the ‘original Markov image set’ template. At the same time, the corresponding the Simpson index is calculated by the DBSCAN clustering network, and the diversity loss function is obtained for back propagation. Legal and appropriate new Markov images are updated to the ‘original Markov image set’, and the ‘new balanced Markov image set’ is output.

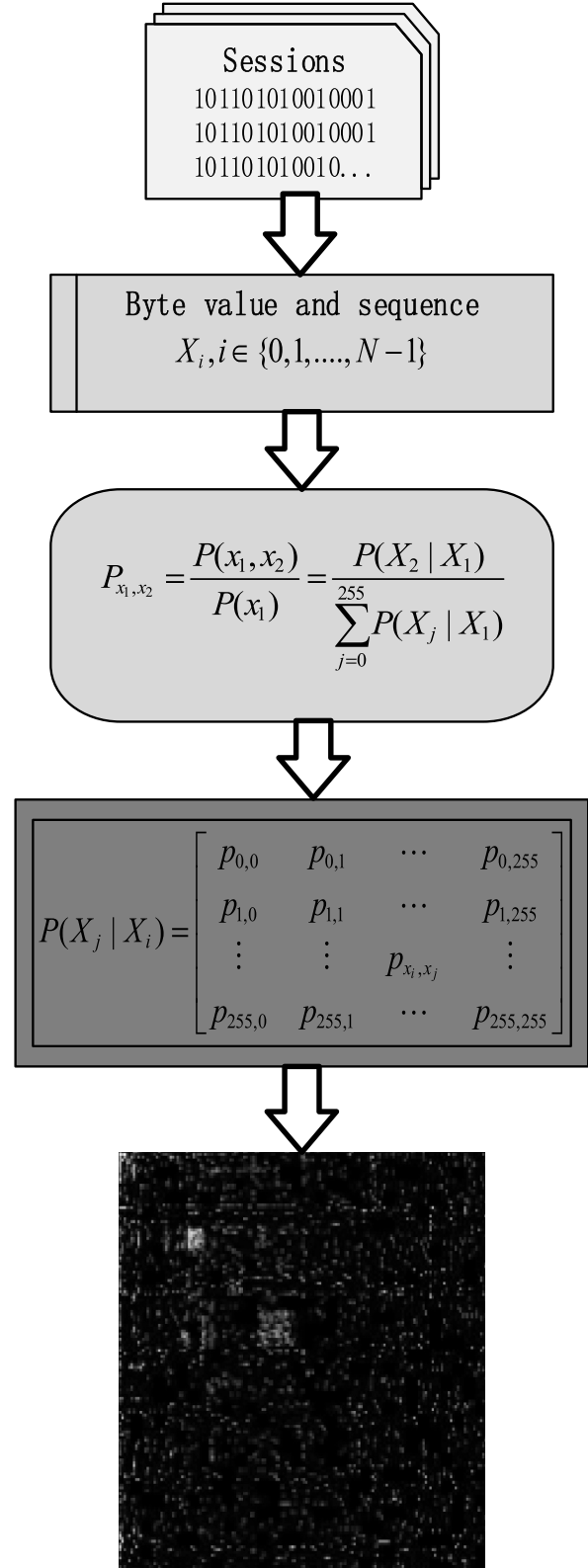


FIGURE 3 Schematic diagram of traffic conversion to the Markov image

In addition, in order to balance the fidelity and diversity of the generated samples, the Markov-GAN abandoned the traditional cross entropy loss function and replaced it with a

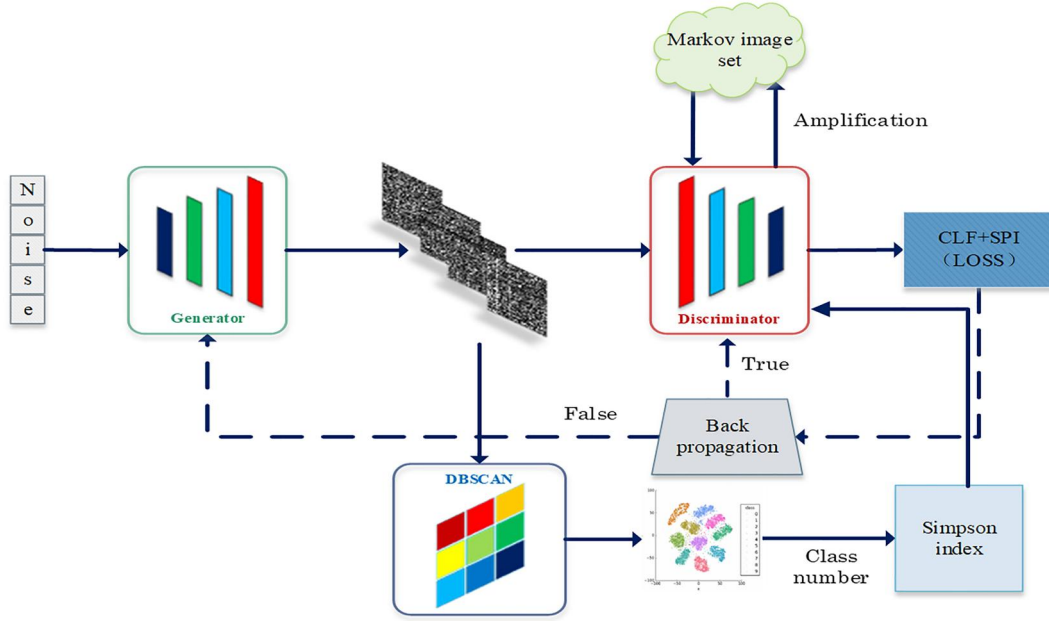


FIGURE 4 Framework of Markov-GAN

classification oriented diversity maximisation loss function and obtained stronger countermeasure samples by designing regularisation constraints. Specific improvements are as follows.

3.2.2 | Diversity maximisation loss function based on compression coding

The Markov-GAN needs to finely control the trade-off between sample fidelity and diversity. The traditional GAN usually measures loss by maximising similarity, but it is difficult to define similarity when it is applied to data with high-dimensional degraded distribution such as encrypted traffic. The classification and clustering task is a data compression mechanism, that is, to find the low rank structure of high-dimensional targets [37]. The generalisation ability can also be regarded as the information compression ability of the classification model to some extent [24]. The GAN network can be well adapted to such compression scenarios, but the generated samples have the problem of homogeneous pattern collapse. Therefore, a new measurement parameter and loss function are designed for classification task, that is, the coding length loss function with the Simpson index as regularisation term. The loss functions of the Markov-GAN discriminator and generator are shown in formulae (4) and (5), respectively.

$$L_{d-loss} = E_{x \sim P_{Tag}} L(x, \epsilon) - E_{x \sim P_g} L(x, \epsilon) - \lambda [SPI - \alpha]^2 \quad (4)$$

$$L_{g-loss} = E_{x \sim P_{Tag}} L(x, \epsilon) - \lambda [SPI - \alpha]^2 \quad (5)$$

In the above formulae, the first item on the right of the equal sign ensures that the generated samples are as real as possible with high fidelity. The second is to ensure that the generated samples have different characteristics as far as possible, that is, the greatest diversity.

Coding length loss function

Due to the strong randomness of the conversion of malicious encrypted traffic file bytes in high-dimensional space, the converted Markov image texture is special and difficult to be accurately quantified by the cross-entropy loss function.

In order to ensure the fidelity of the generated samples, considering the physical significance of the transition probability of adjacent fields, a coding length function (CLF) is proposed to measure the degree of information compression. This function has the ability of image compression representation and can guide the Markov-GAN to generate new Markov images with family homology and texture similarity, so as to improve the game level between the discriminator and generator. The coding length function is a data compression representation method proposed in combination with rate distortion theory, as shown in formula (6):

$$L(x, \epsilon) = \left(\frac{m+D}{2} \right) \log \det \left(I + \frac{D}{m\epsilon^2} XX^T \right) \quad (6)$$

Where m and D are the size of the data, respectively, corresponding to the length and width of the Markov image. I is the upper network residual, and X is the image tensor. ϵ is a constraint constant parameter used to determine the compression accuracy, which is generally 0.000,001. The whole coding length function can maximise the space occupied by its expression while making the samples of the same structure

close, that is, the sample expression ability is the largest. At the same time, when the sample expression between different structure data is optimal, the overall sample set has the maximum coding length.

Regular term based on the Simpson index

The Simpson index is a biological index used to measure species diversity in communities [39]. It is used here to evaluate the species richness of the generated Markov images. In order to enhance the diversity of generated samples and maximise the range of each type of data, the Simpson index is taken as the regular term of loss function, as shown in equation (7).

$$SPI = 1 - \sum_{i=1}^n P_i \quad (7)$$

Where, P_i represents the probability that a class is selected in an encrypted traffic dataset, and its calculation method is shown in formula (8).

$$P_i = \frac{\text{Number of individuals in a group}}{\text{Total number of individuals in the community}} \quad (8)$$

The entire loss function is back propagated by initialising the α parameter (generally no more than 1), and the SPI regularisation constraint is made to make the value as close to α as possible, so as to guide the generator G to generate the Markov image with the maximum type to achieve enhancement and balance the goal.

3.2.3 | Training algorithm of the Markov-GAN

The generator G and the discriminator D of the Markov-GAN use a 5-layer deconvolutional neural network and a 5-layer convolutional neural network, respectively. The first four layers of generator G use the Leaky-Relu function as the activation function, and the last layer uses the tanh activation function. The discriminator D uses the Leaky-Relu function as its activation function, and its output layer is a Dense layer with an output node of 1 to judge the 'true' and 'false' of the generated Markov image. Based on the results of DBSCAN clustering, the Simpson index and the corresponding loss function are calculated. At the same time, the Adam optimiser is used to update the model parameters and weights by alternately training discriminator D and generator G until Nash equilibrium is reached. The specific training process of Markov-GAN is shown in algorithm 2.

Algorithm 2 Markov-GAN, our proposed algorithm

- **Input:** learning rate (α) = 0.01; beta_1 = 0.9; beta_2 = 0.999; batch size (m);
- **Output:** generator parameters (θ_g); discriminator parameters (θ_d); Classifier

```

parameters ( $C$ ); Simpson index ( $spi$ ); Height
and width of data ( $D, m$ ); Control parameters
( $\varepsilon$ ); Constant ( $\theta$ ); Identity ( $I$ );
1: while  $\theta_g$  has not converged do
2: For  $I = 1, \dots, m$  Do
3: Markov image  $x \sim P_{data}(x)$ ; Noise sample
 $z \sim P(z)$ ;
4: Interpolation sampling of samples  $G(z)$ ;
5: Coding Length Function ( $CLF$ )  $L(x, \varepsilon) =$ 
 $(D + m/2) \log \det(I + D/m\varepsilon^2 XX^T)$ 
6: Loss function of generator and
discriminator: // Use CLF and SPI to realise
the GAN loss function
7:  $L_{d\_loss} = E_{x \sim P_{data}} L(x, \varepsilon) - E_{x \sim P_g} L(x, \varepsilon)$ 
 $-\lambda[spi - \theta]^2$ 
8:  $L_{g\_loss} = E_{x \sim P_g} L(x, \varepsilon) - \lambda[spi - \theta]^2$ 
9: Update discriminator parameters  $\theta_d$  and
generator parameters  $\theta_g$ :
10: While  $D$  and  $G$  doesn't converge do
11:   While  $C$  clustering is complete do
12:      $spi \leftarrow spi(C(G(z)), \theta)$  // Simpson
index is calculated by clustering network
13:      $\theta_d \leftarrow \alpha \cdot Adam(b_1, b_2, \Delta \theta_d(L_{d\_loss}))$ 
14:   Then
15:      $\theta_g \leftarrow \alpha \cdot Adam(b_1, b_2, \Delta \theta_g(L_{g\_loss}))$ 
16:   End
17: End
18: End While

```

4 | EXPERIMENTAL EVALUATION

4.1 | The datasets and evaluation indicators

4.1.1 | Original dataset

The experiment was conducted on several typical original encrypted datasets, as shown in Table 4.1, Table 4.2, and Table 4.3 for details. These datasets cover multiple types of encrypted traffic, and all have the problem of imbalanced class distribution, which can be used to test the enhancement ability, generalisation ability and classification accuracy of the proposed model. The first is the USTC-TFC2016 (Part I) dataset [21], which contains 10 kinds of malware traffic from public websites collected from real network environment from 2011 to 2015 [40]. The proportion of various samples in the dataset is extremely uneven. For example, the number of sessions of Htbot and Tinba malicious encryption traffic accounts for only 2.78% and 3.71%, respectively, while Cridex and Geodo account for 26.24% and 18.10%, respectively. The second is the USTC-TFC2016 dataset (Part II), which contains 10 types of normal traffic and can be used for encrypted traffic classification of different application types. Among them, there are only 2990 Facetime streams and 11,506 FTP streams, which vary greatly. The third is the ISCXVPN2016 dataset [10], including 15 different VPN encryption applications. The number of samples

in many categories is far lower than the average, and the degree of imbalance is very large. Based on these original public datasets, three kinds of experiments are given, including enhancement and balance ability experiment, accuracy comparison experiment and representation and generalisation ability experiment.

4.1.2 | Performance metrics

In order to evaluate the performance indicators under imbalanced datasets from multiple dimensions, a set of performance indicators are used for comprehensive evaluation, including accuracy, precision, recall and F1 score, which are defined as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

$$\text{F1_Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

Among them, TP and FP are the number of samples correctly and incorrectly classified as wrong, respectively. Correspondingly, TN and FN are the number of samples correctly and incorrectly classified as correct, respectively.

4.1.3 | Experimental platform and model structure parameters

The proposed model is trained, tested and used in the Windows environment. The configuration parameters are Intel I7-10,875H CPU 5.1 GHz, 32 GB RAM and external GPU (NVIDIA GeForce RTX 2070S). The relevant code is developed in Python language based on TensorFlow 2.0 framework.

The parameters of the Markov-GAN and CNN models used in this paper are shown in Table 2 and Table 3, respectively. DBSCAN Clustering algorithm's eps is 2.4 and min_samples is 20.

4.2 | Experimental results and analysis

4.2.1 | Dataset sample enhancement and balance experiment

Data set conversion and balance.

TABLE 2 Markov-GAN structure and parameters

Markov-GAN	Layer	Activation	Shape
Generator	Input		96
	Dense		1024
	Dense		2048
	Dense	Tanh	65,536
Discriminator	Input		(256,256)
	Dense	Relu	512
	Dense		256
	Dense		1

Note: (1) Other parameters: batch_size = 128; learn_rate = 0.001; epochs = 100, (2) Generator Optimisation algorithm parameters: optimiser:Adam, beta_1 = 0.9; beta_2 = 0.999, (3) Discriminator Optimisation algorithm parameters: optimiser:Adam, beta_1 = 0.9; beta_2 = 0.999, and (4) input noise size: 256 × 256.

TABLE 3 CNN structure and parameters

Model	Layer	Rate	Activation	Filters	F · size	Strides
CNN	Conv 0		Relu	20	5	(2,2)
	Pool 0				(2,2)	(2,2)
	Conv1		Relu	40	4	(2,2)
	Pool 1				(2,2)	(2,2)
	Flatten					
	Dense		Relu			
	Dropout	0.3				
	Dense		Arg_max			

Note: (1) Other parameters: batch_size = 128; learning_rate = 0.001; epochs = 50 and (2) Optimisation algorithm parameters: optimiser: Adam, beta_1 = 0.9; beta_2 = 0.999.

First, the original encrypted traffic dataset in the form of flow is transformed into a session form, and the traffic is converted into Markov image one-to-one in the unit of session to obtain the 'original Markov image set'. The image set was used as validation and test set in subsequent experiments. Second, according to the imbalance of the original dataset, the weak samples of the 'original Markov image set' are generated, amplified and balanced based on the Markov-GAN to obtain a 'new balanced Markov image set', which is used as a training set in subsequent experiments. Table 4, Table 5 and Table 6 show the situation before and after the balance of the three datasets.

It is worth noting that the proposed method balances the converted Markov image set rather than the traffic of the original data set. In addition, the Markov-GAN has a strong generation ability, which can be amplified and balanced to varying degrees in different proportions. Limited to space, only the minimum equilibrium proportion when the accuracy improvement reaches saturation under experimental conditions is given in the tables.

TABLE 4 USTC-TFC2016 (Part I) data generation and balance

Malicious type	Original dataset			New balanced Markov image set	
	Flow	Session	Proportion of sessions	Image	Proportion of image
Cridex	90,087	60,058	26.24%	15,000	10.00%
Geodo	42,087	41,418	18.10%	15,000	10.00%
Htbot	12,734	6367	2.78%	15,000	10.00%
Miuref	26,962	13,481	5.89%	15,000	10.00%
Neris	69,825	36,751	16.59%	15,000	10.00%
Nsis-ay	12,138	6069	2.65%	15,000	10.00%
Shifu	19,268	9634	4.21%	15,000	10.00%
Tinba	17,008	8504	3.71%	15,000	10.00%
Virut	71,200	35,600	15.56%	15,000	10.00%
Zeus	21,940	10,970	4.79%	15,000	10.00%
Total	383,249	228,852	100%	150,000	100%

TABLE 5 USTC-TFC2016 (Part II) data generation and balance

Application type	Original dataset			New balanced Markov image set	
	Flow	Session	Proportion of sessions	Image	Proportion of image
BitTorrent	7535	3767	16.03%	10,000	10.00%
Facetime	2990	2990	6.76%	10,000	10.00%
FTP	11,506	5753	13.01%	10,000	10.00%
Gmail	11,477	5723	12.95%	10,000	10.00%
MySQL	11,385	5692	12.86%	10,000	10.00%
Outlook	7467	3733	8.44%	10,000	10.00%
Skype	6028	3014	6.81%	10,000	10.00%
SMB	11,543	5770	13.05%	10,000	10.00%
Weibo	11,510	5755	13.02%	10,000	10.00%
WorldOfWarcraft	11,559	5779	13.07%	10,000	10.00%
Total	93,000	44,209	100%	100,000	100%

4.3 | Experiment on generation quality and diversity of Markov images

In order to test the texture quality and diversity of the new Markov image generated by the Markov-GAN, based on the USTC-TFC2016 (Part I) dataset, the comparison between the converted part of the original Markov image and the generated Markov image is given. Figure 5a shows the original Markov image of Cridex class, and b,c,d are the generated new Markov images. Figure 6 shows different types of Markov images generated.

It can be seen that, on the one hand, the Markov images generated in the same class have similar texture distribution with the original Markov images, that is, they have intra-class similarity and diversity. On the other hand, the generated Markov images of different classes have significant inter-class differences, and their texture feature distributions are very

different. It can be seen that the coding loss function of the Markov-GAN and Simpson diversity index can work effectively. While ensuring the good fidelity of the generated image, it can enhance the diversity and greatly expand the number of effective samples.

5 | CLASSIFICATION ACCURACY AND COMPARATIVE EXPERIMENT

5.1 | Effects of different imbalance degrees and generalisation degrees on accuracy

In order to further explore the relationship between the amount of generated samples and the imbalance degree of dataset and classification accuracy, two quantitative parameters, imbalance degree and generalisation degree, are proposed.

Application type	Original dataset			New balancedMarkov image set	
	Flow	Session	Proportion of sessions	Image	Proportion of image
AIM_chat	4869	4823	4.29%	10,000	6.67%
Email	4417	4417	3.93%	10,000	6.67%
Facebook	5527	2763	2.46%	10,000	6.67%
Gmail	7329	3664	3.26%	10,000	6.67%
Hangouts	7587	3793	3.37%	10,000	6.67%
ICQ	4243	3946	3.51%	10,000	6.67%
Netflix	51,932	15,966	14.19%	10,000	6.67%
SCPdown	15,390	7695	6.84%	10,000	6.67%
SFTPdown	4729	2364	2.11%	10,000	6.67%
Skype	4607	2303	2.05%	10,000	6.67%
Spotify	14,442	7221	6.42%	10,000	6.67%
TorTwitter	14,654	7327	6.51%	10,000	6.67%
Vimeo	18,755	9377	8.34%	10,000	6.67%
Voipbuster	35,469	29,456	26.19%	10,000	6.67%
Youtube	12,738	7369	6.55%	10,000	6.67%
Total	206,688	112,484	100%	150,000	100%

TABLE 6 ISCXVPN2016 data generation and balance

The imbalance degree is defined as the average of the absolute value of the difference between the size of each type of sample and the average of the total sample size, as shown in Equation (13):

$$UD = \frac{\sum |X_i - \mu|}{N} \quad (13)$$

Where, UD is the imbalance degree, X_i is the number of samples of each class, μ is the average value of all samples, and N is the total number of samples.

Generalisation degree is defined as the ratio of the total number of samples (including converted and generated) after balance to the converted original sample size, as shown in Equation (14):

$$GD = \frac{Original + Generated}{Original} \quad (14)$$

Where GD is the generalisation degree, *Original* is the original sample size obtained by conversion, and *Generated* is the generated sample size.

The experiment was conducted in three dimensions: imbalance degree, generalisation degree and accuracy. Based on the USTC-TFC2016 (Part I) dataset, the number of samples of the most vulnerable category is taken as the generalisation base, that is, 5000 only. At the same time, five imbalance degrees are artificially set, which are 0%, 20%, 40%, 60% and 80%. For example, 0% means that the dataset is balanced, that is, 500 samples for each class. Then, the generated quantity is

generalised to different degrees while keeping the imbalance degree unchanged. Finally, the converted 'original Markov image set' is used for verification and test, and their classification accuracy is counted, respectively. The results are shown in Figure 7.

It can be seen that under the same total sample size, the classification accuracy increases gradually with the imbalance degree from high to low, which indicates that the balanced dataset improves the classification effect. At the same time, under the same imbalance degree, the accuracy will increase significantly with the increase of generalisation degree of generated samples, and the optimal accuracy all exceed 90%. It should be noted that the improvement of accuracy is saturated, that is, when the generalisation degree increases to a certain extent, the classification effect will not be improved. This critical point can be used to guide the selection of generalisation degree, which is also the basis for selecting corresponding balance quantities in Table 4.1, Table 4.2 and Table 4.3. The experimental results show that by indirectly enhancing and balancing datasets, the Markov-GAN can be applied to datasets of multiple types and multiple degrees of imbalance. While significantly improving the classification accuracy, it greatly reduces the dependence and requirements on the quantity and quality of original datasets. A small number of original samples can make a good prediction of the whole dataset.

5.2 | Comparison with other data enhancement methods

In order to compare the data enhancement and balance effects of different generation models, the accuracy rate, recall rate

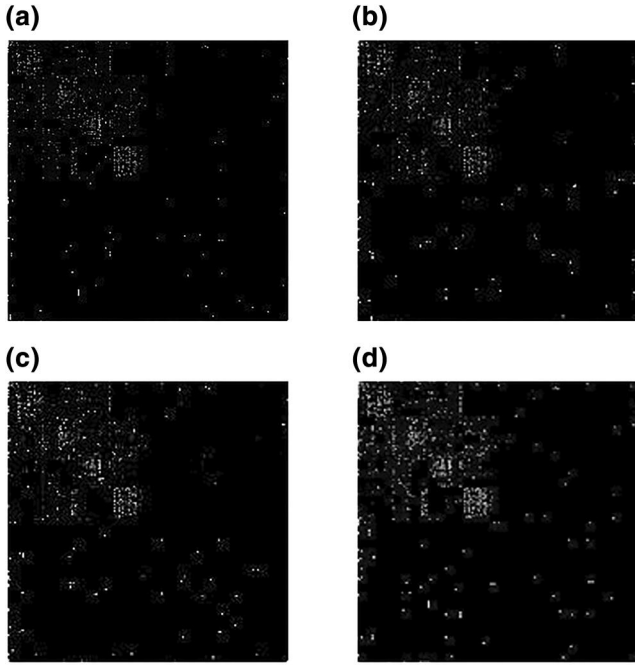


FIGURE 5 Markov image sample of Cridex class

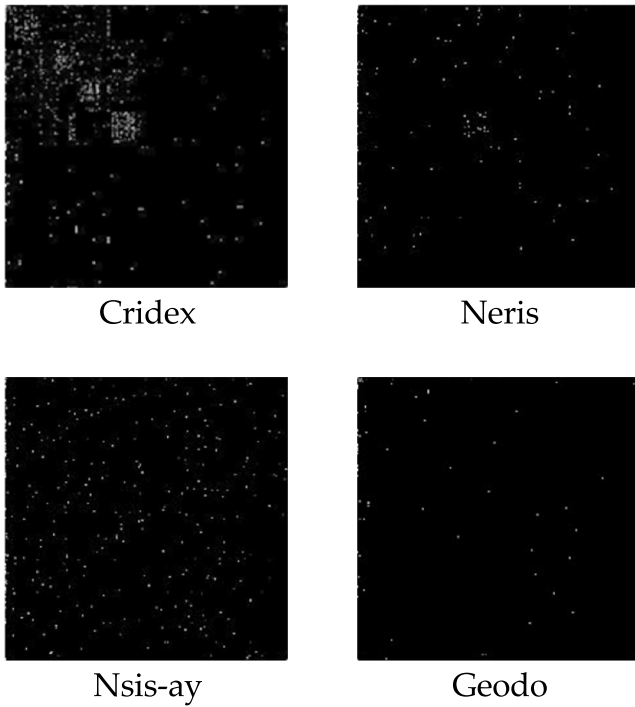


FIGURE 6 Markov image samples of different classes

and F1 score are compared with the FLOWGAN [23]. Different from the Markov-GAN, the FLOWGAN generates encrypted traffic and expands and balances the original traffic dataset. Based on ISCXPVN2016 and the dataset enhanced by the FLOWGAN and Markov-GAN, respectively, the experimental data of corresponding classification effect are shown in Table 7.

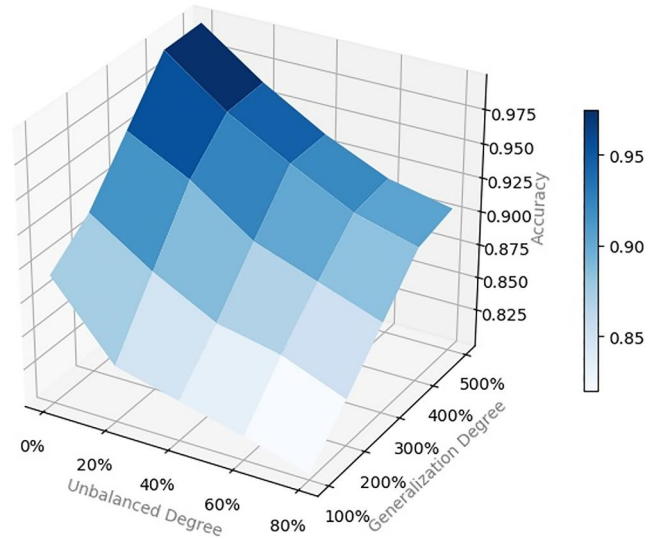


FIGURE 7 Influence of imbalance and generalisation degree on accuracy

It can be seen that the weak class of the original dataset before balancing, such as AIM_chat and voipbuster, have poor classification results. After using the GAN method for data enhancement, the classification rates of all 15 classes have increased to a high level, especially the weak class. For example, all corresponding indicators of the AIM_chat category have risen by nearly 30%. Compared with the FLOWGAN, the average values of the three indicators of the Markov-GAN have further increased, all exceed 99%. The experiment shows that compared with the method of simply expanding the number of samples, the Markov-GAN also considers the fidelity and diversity of the generated samples, which makes the classification effect reach a higher level.

In order to further compare the ability of different data enhancement methods to improve classification accuracy, the related work is divided into three groups: original sample balance based on artificial, machine-based original sample generation and balancing and machine-based transform domain sample generation and balance. The experimental results are shown in Table 8.

It can be seen from the table that compared with other classification methods that do not use enhancement techniques, such as feature extraction-based, cost-sensitive, and image transformation-based classification techniques, the method proposed in this paper using the Markov-GAN for data augmentation and then classification is more effective. Compared with the other two data enhancement methods, the accuracy and accuracy growth obtained by the Markov-GAN are higher. More importantly, the classification effect of the ordinary CNN is equivalent to that of complex models such as VGG13 and VGG16, all of which are maintained at more than 99.6%. It shows that the image set enhanced by the Markov-GAN is more friendly to the model and can effectively alleviate the performance generalisation deviation caused by different network depths.

TABLE 7 Comparison of classification results of the original dataset, FLOWGAN dataset, and Markov-GAN dataset

Class	Original dataset			FLOWGAN dataset			Markov-GAN dataset		
	Precision	Recall	F1_Score	Precision	Recall	F1_Score	Precision	Recall	F1_Score
AIM_chat	0.6987	0.5558	0.6191	0.9818	0.9496	0.9654	0.9913	0.9604	0.9756
Email	0.9655	0.9801	0.9728	1.0000	0.9930	0.9965	0.9944	0.9921	0.9932
Facebook	0.8313	0.6698	0.7419	0.9883	0.9798	0.9840	0.9893	0.9801	0.9847
Gmail	0.9392	0.9245	0.9318	0.9821	0.9860	0.9840	0.9871	0.9868	0.9869
Hangouts	0.6915	0.9282	0.7926	0.9879	0.9911	0.9895	0.9892	0.9910	0.9900
ICQ	0.7673	0.7280	0.7471	0.9394	0.9876	0.9629	0.9916	0.9965	0.9940
Netfilx	0.9734	0.9973	0.9852	0.9965	0.9938	0.9951	0.9946	0.9971	0.9952
Scpdown	0.9721	0.9997	0.9857	0.9997	1.0000	0.9998	1.0000	0.9996	0.9992
Sftpdown	0.9647	0.8299	0.8922	0.9978	0.9983	0.9980	0.9976	0.9982	0.9979
Skype	0.8949	0.8937	0.8943	0.9957	0.9922	0.9939	0.9889	0.9931	0.9910
Spotify	0.912 4	0.9866	0.9480	0.9970	0.9882	0.9925	0.9923	0.9934	0.9928
torTwitter	0.9934	0.9993	0.9963	0.9985	0.9993	0.9989	1.0000	0.9987	0.9993
Vimeo	0.9650	0.9945	0.9795	0.9852	0.9963	0.9894	0.9899	0.9946	0.9982
Voipbuster	0.8967	0.9928	0.9423	0.9982	0.9905	0.9943	0.9942	0.9956	0.9949
Youtube	0.9566	0.9997	0.9777	0.9982	0.9995	0.9988	0.9987	1.0000	0.9993
Average	0.8948	0.8987	0.8938	0.9898	0.9897	0.9895	0.9933	0.9918	0.9925

TABLE 8 Comparison with other methods

Technical route	Paper	Classification method	Balance model	Accuracy	Accuracy growth
Unenhanced classification methods	[42]	Multilayer perceptron	———	0.9372	———
	[43]	CSCNN	———	0.9790	———
	[21]	CNN	———	0.9917	———
Original sample balance-based on artificial	[18]	CNN	Semi-supervised	0.9850	0.0040
	[20]	CNN	ROS	0.9889	0.0092
	[20]	CNN	Smote	0.9769	−0.0028
Machine-based original sample generation and balancing	[20]	CNN	PacketCGAN	0.9951	0.0154
	[8]	RF	AC-GAN	0.9989	0.0011
	[23]	MLP	FLOWGAN	0.9910	0.0915
Machine-based transform domain sample generation and balance	Ours	CNN	Markov-GAN	0.9960	0.0965
	Ours	VGG13 [41]	Markov-GAN	0.9975	0.0980
	Ours	VGG16 [41]	Markov-GAN	0.9962	0.0967

5.3 | REPRESENTATION AND GENERALISATION ABILITY EXPERIMENT

5.3.1 | Representation ability experiment of different transform domain methods

In order to test the performance of different representation learning methods of encrypted traffic, the Markov image and grey image are used for comparison test under the same

classification model. Based on USTC-TFC2016 (Part I), 1000 Markov images of each type generated and balanced are used as training set samples, and 100 images of each type are extracted from the converted ‘original Markov image set’ as test set samples. Accordingly, an equal amount of grey image is converted, generated and balanced. Several comparison results based on the two images are obtained in the experiment. The confusion matrix of classification is shown in Figure 8, and the three indexes of precision, recall and F1 score are shown in Table 9 and Figure 9, respectively.

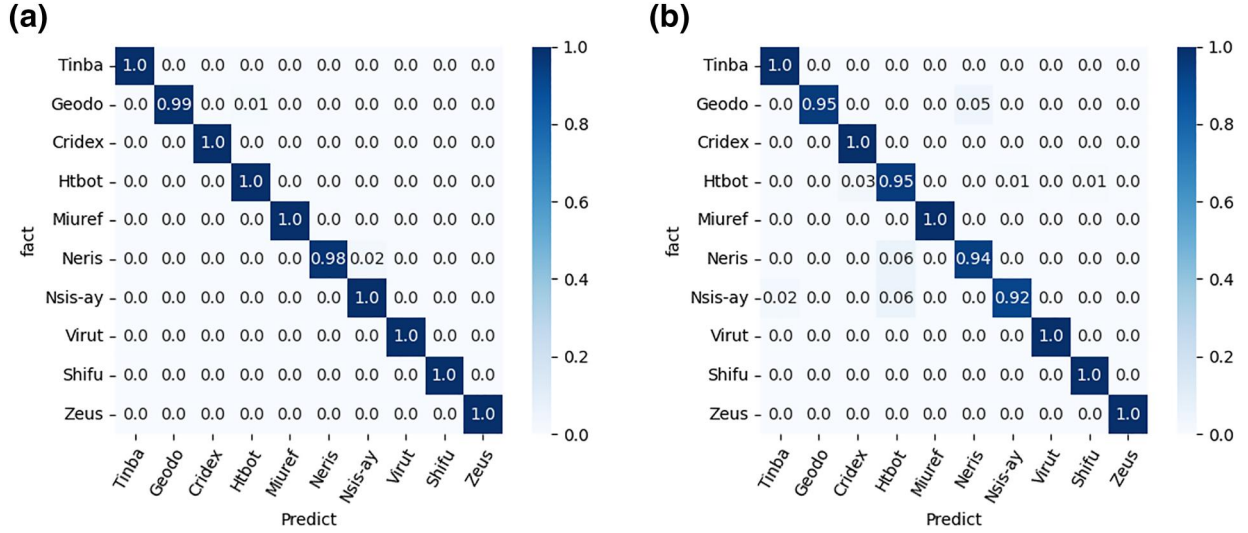


FIGURE 8 Confusion matrix based on Markov image and grey image (a) Method based on Markov image and (b) method based on grey image

TABLE 9 Comparison of precision, recall and F1 score of two transform domain methods

Malicious type	Method based on Markov image			Method based on grey image		
	Precision	Recall	F1 score	Precision	Recall	F1 score
Cridex	1	1	1	0.970	1	0.985
Geodo	0.990	0.990	0.990	0.950	0.950	0.950
Htbot	0.990	1	1	0.888	0.950	0.918
Miuref	1	1	1	1	1	1
Neris	0.980	0.980	0.980	0.949	0.940	0.944
Nsis-ay	0.980	1	0.990	0.989	0.920	0.953
Shifu	1	1	1	0.990	1	0.995
Tinba	1	1	1	0.980	1	0.990
Virut	1	1	1	1	1	1
Zeus	1	1	1	1	1	1
Average value	0.994	0.997	0.996	0.972	0.976	0.974

As can be seen from Figure 8, both the method based on the Markov image and the method based on the grey image have high classification accuracy. In contrast, the former has better results, with eight out of 10 categories achieving 100% classification accuracy. Table 9 and Figure 9 give detailed evidence from three indicators of precision, recall and F1 score. It shows that the Markov image not only has no preprocessing distortion such as traffic slicing and filling, but also integrates more feature information such as the timing relationship between adjacent bytes, so it has stronger representation ability.

5.3.2 | Performance comparison of different transform domain characterisation methods

10,000 Pcap traffic packets of different sizes are randomly selected from the three datasets and converted into Markov

images and grey images, respectively. The statistical results of image file size are shown in Table 10.

It can be seen that compared with the grey image, the maximum decrease of the Markov image is more than 70%, the maximum is 98%, and it is always maintained at a low level. The size of grey image will increase with the increase of traffic, and the picture specification cannot be naturally unified. Markov image conversion does not need to slice and fill the traffic artificially, and the image specifications are naturally unified (256×256), so it has good universality and can be expressed uniformly across datasets.

5.3.3 | Migration experiments across different datasets

In order to evaluate the portability of the proposed method, classification experiments were carried out on the CNN using

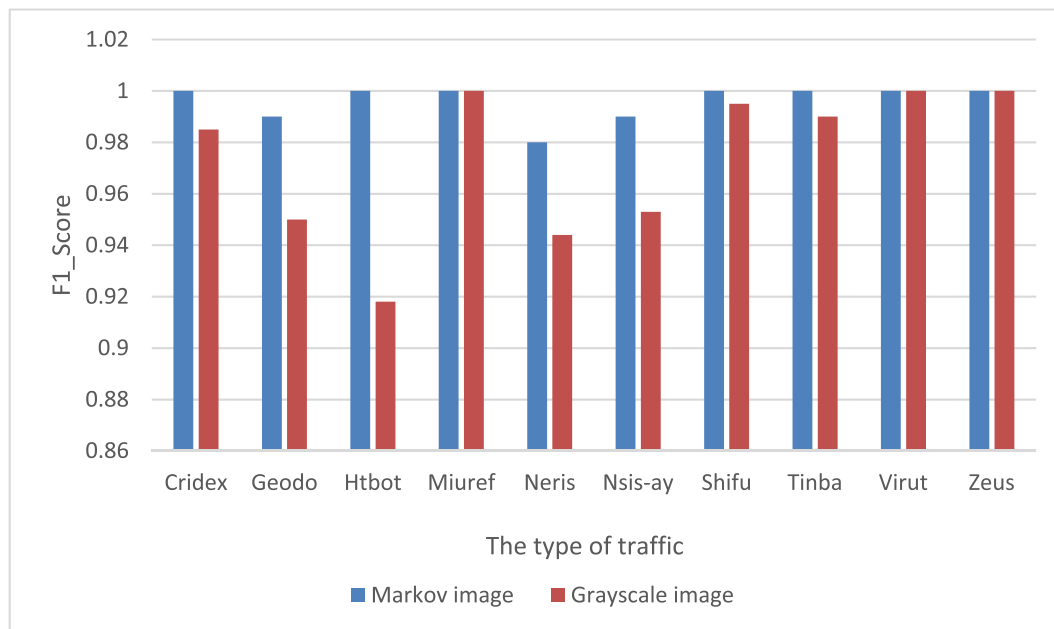


FIGURE 9 Comparison of F1 score of two transform domain methods

TABLE 10 statistical comparison of Markov image and grey image size

Dataset	Flow size	Markov image size	Grey image size	Maximum decrease
USTC-TFC2016 (Part I)	1–10 KB	3.37–18.2 KB	2.02–6.48 KB	97.5%
	10–100 KB	12.7–18.2 KB	6.48–63 KB	
	100–925 KB	6.25–12.7 KB	63–517 KB	
	925 KB+	—	—	
USTC-TFC2016 (Part II)	1–10 KB	4.2–17 KB	2.21–5.97 KB	70.7%
	10–95 KB	10.8–17 KB	5.97–58.1 KB	
	95 KB+	—	—	
ISCXVPN2016	1–10kB	4.83–18.5 KB	1.98–5.06 KB	98.1%
	10–100kB	11.2–18.5 KB	5.06–63 KB	
	100–1000 KB	5.67–11.2 KB	63–616 KB	
	1000 KB+	5.67–11.2 KB	616 KB+	

the balanced datasets in Table 4, Table 5 and Table 6. The average values of precision, recall and F1 score are shown in Table 11. The data show that the proposed method not only achieves high scores on three different datasets but also has very little performance difference between them. It shows that the transformation domain enhancement and balance method based on the Markov-GAN has good generalisation ability for heterogeneous datasets.

6 | CONCLUSION

The quantity and quality problems of malicious encrypted traffic datasets result in the low accuracy and weak generalisation ability of the deep learning classification model. In

TABLE 11 Performance of the Markov-GAN method in different datasets

Dataset	Precision	Recall	F1_Score
USTC-TFC2016 (part I)	0.996	0.997	0.996
USTC-TFC2016 (part II)	0.982	0.964	0.972
ISCXVPN2016	0.9933	0.9918	0.9925

order to solve the classification problem under the condition of imbalanced class distribution, two studies are done in this paper. On the one hand, a new encrypted traffic representation learning is proposed, that is, the image diversity of low rank feature space is used to represent the sample diversity of

high-dimensional traffic space, which realised the unified expression across datasets. Experiments show that compared with grey image and other transform domain methods, the Markov image has stronger representation ability, lighter weight and smaller size and is more friendly to the model. On the other hand, in order to balance the Markov image dataset, a Markov-GAN enhancement model is proposed, which can automatically generate new Markov images with rich texture, high discrimination and good diversity. Experiments show that compared with other dataset balancing methods, we obtain higher classification accuracy with less original sample size by balancing Markov image sets, and all of them exceed 90%. Moreover, for different datasets and different depth learning networks, the performance generalisation deviation of the proposed method is greatly reduced. Compared with other methods that directly enhance the original traffic samples, such as PacketCGAN and AC-GAN, the Markov-GAN is not only lighter and easier to train but also has stronger sample amplification ability and generalisation ability.

ACKNOWLEDGEMENTS

This work was supported in part by the National Key Research and Development Program under Grant 2018YFB0804503 and Grant 2019QY1400, in part by the National Natural Science Foundation of China under Grant U20A20161 and Grant U1836103, and in part by the Basic Research Program of China under Grant 2019-JCJQ-ZD-113.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

PERMISSION TO REPRODUCE MATERIALS FROM OTHER SOURCES

None.

ORCID

Zhangguo Tang  <https://orcid.org/0000-0001-6405-8445>

REFERENCES

- Rezaei, S., Liu, X.: Deep learning for encrypted traffic classification: an overview. *IEEE Commun. Mag.* 57(5), 76–81 (2019)
- Cisco, G.S.: ETA – Provides Solution for Detecting Malware in Encrypted Traffic [EB]. 2018-01-14/2021-07-16 <http://gbhackers.com/cisco-eta-encrypted-traffic/>
- Cisco, C.: Encrypted Traffic Analytics White paper[R/OL]. (2021-05)[2021-05-20] <https://www.cisco.com/c/en/us/solutions/collateral/enterprise-networks/enterprise-network-security/nb-09-encrytd-traf-anlytcs-wp-cte-en.pdf>
- Security, H.N.: Encryption-based Threats Grow by 260% in 2020[R/OL]. (2020-11-11)[2021-06-19] <https://www.helpnetsecurity.com/2020/11/11/encryption-based-threats-grow-2020/>
- Magnúsdóttir, A.: Malware Is Moving Heavily to HTTPS [EB]. <https://www.cyren.com/blog/articles/over-one-third-of-malware-uses-https>, 2017-06-07/2021-04-16
- He, G., et al.: Mobile app identification for encrypted network flows by traffic correlation. *Int. J. Distributed Sens. Netw.* 14(12), 1550147718817292 (2018). <https://doi.org/10.1177/1550147718817292>
- Meghdouri, F., Zseby, T., Iglesias, F.: Analysis of lightweight feature vectors for attack detection in network traffic. *Appl. Sci.* 8(11), 2196 (2018). <https://doi.org/10.3390/app8112196>
- Vu, L., Bui, C.T., Nguyen, Q.U.: A deep learning based method for handling imbalanced problem in network traffic classification. In: *Proceedings of the Eighth International Symposium on Information and Communication Technology*, pp. 333–339 (2017)
- Cheng, H., Xie, J., Chen, L.: CNN-based encrypted C&C communication traffic identification method. *Comput. Eng.* (2019)
- Lashkari, A.H., et al.: Characterization of encrypted and VPN traffic using time-related features. In: *The International Conference on Information Systems Security and Privacy*. (ICISSP) (2016)
- Torroledo, I., Camacho, L.D., Bahnsen, A.C.: Hunting malicious TLS certificates with deep neural networks. In: *Proceedings of the 11th ACM Workshop on Artificial Intelligence and Security*, pp. 64–73 (2018)
- Zeng, Y., et al.: Research on malicious traffic identification technology in encrypted traffic (in Chinese). *J. Xidian Univ.* 48(03), 170–187 (2021)
- Guo, L., et al.: Deep learning-based real-time VPN encrypted traffic identification methods. *J. R. Time Image. Process.* 17(1), 103–114 (2020)
- Meng, S., et al.: Classification of encrypted traffic with second-order Markov chains and application attribute bigrams. *IEEE Trans. Inf. Forensics Secur.* 12(8), 1830–1843. <https://doi.org/10.1109/tifs.2017.2692682>, (2017)
- wang, R.H., et al.: An LSTM-based deep learning approach for classifying malicious traffic at the packet level. *Appl. Sci.* 9(16), 3414 (2019). <https://doi.org/10.3390/app9163414>
- Wei, W., et al.: End-to-end encrypted traffic classification with one-dimensional convolution neural networks. In: *IEEE International Conference on Intelligence and Security Informatics*. (ISI) (2017)
- Alan, H.F., Kaur, J.: Can android applications Be identified using only TCP/IP headers of their launch time traffic? In: *Acm Conference on Security & Privacy in Wireless & Mobile Networks*, 61–66 (2016)
- Ilyasu, A.S., Deng, H.: Semi-supervised encrypted traffic classification with deep convolutional generative adversarial networks. *IEEE Access.* 8, 118–126 (2020). <https://doi.org/10.1109/access.2019.2962106>
- Rezaei, S., Liu, X.: How to Achieve High Classification Accuracy with Just a Few Labels: A Semi-supervised Approach Using Sampled Packets (2018)
- Wang, P., et al.: PacketCGAN: Exploratory Study of Class Imbalance for Encrypted Traffic Classification Using CGAN (2019)
- Wei, W., et al.: Malware traffic classification using convolutional neural network for representation learning. In: *International Conference on Information Networking*. (ICOIN) (2017)
- Korczynski, M., Duda, A.: Markov chain fingerprinting to classify encrypted traffic. In: *IEEE INFOCOM 2014-IEEE Conference on Computer Communications*, pp. 781–789. *IEEE* (2014)
- Wang, Z.X., et al.: FLOWGAN: Unbalanced network encrypted traffic identification method based on GAN. In: *IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking*. (ISPA/BDCloud/SocialCom/SustainCom) (2019)
- Lu, X., et al.: Dynamics Generalization via Information Bottleneck in Deep Reinforcement Learning. *arXiv preprint arXiv:2008.00614* (2020)
- Lotfollahi, M., et al.: Deep packet: a novel approach for encrypted traffic classification using deep learning. *Soft Comput.* (3), 1999–2012 (2017). <https://doi.org/10.1007/s00500-019-04030-2>
- Bazuhair, W., Lee, W.: Detecting malign encrypted network traffic using Perlin noise and convolutional neural network. In: *2020 10th Annual Computing and Communication Workshop and Conference*. (CCWC) (2020)
- Burnap, P., et al.: Malware classification using self organising feature maps and machine activity data - ScienceDirect. *Comput. Secur.* 73, 399–410 (2018)

28. Wang, S., et al.: TrafficAV: an effective and explainable detection of mobile malware behavior using network traffic. In: IEEE/ACM 24th International Symposium on Quality of Service (IWQoS) (2016)
29. Hu, B., et al.: Malicious traffic detection combining features of packet payload and stream fingerprint (in Chinese). *Comput. Eng.* 46(11), 163–169 (2020)
30. Anderson, B., Paul, S., McGrew, D.: Deciphering malware's use of TLS (without decryption). *J. Comput. Virol. Hacking. Tech.* 14(1), 1–17 (2016)
31. Nychis, G., et al.: An empirical evaluation of entropy-based traffic anomaly detection. In: Proceedings of the 8th ACM SIGCOMM Conference on Internet Measurement, pp. 151–156 (2008)
32. Fang, J.: Research on malicious TLS traffic identification based on hybrid neural network. In: 2020 International Conference on Advance in Ambient Computing and Intelligence. (ICAACI) (2020)
33. Gong, G.L., Yi, J.K., Zhang, Y.C.: Encrypted traffic classification on length-ware constrained clustering (in Chinese). *J. Chongqing Univ. Technol. Nat. Sci.* 35(5), 9 (2021)
34. Yuan, B., et al.: Byte-level malware classification based on markov images and deep learning. *Comput. Secur.* 92, 101740 (2020). <https://doi.org/10.1016/j.cose.2020.101740>
35. Wang, P., Chen, X.: SAE-based Encrypted Traffic Identification Method. *Computer Engineering* (2018)
36. Japkowicz, N.: Learning from imbalanced data sets: a comparison of various strategies. In: Aaai Workshop on Learning from Imbalanced Data Sets. 68, 10–15 (2000)
37. Chawla, N.V., et al.: SMOTE: synthetic minority over-sampling technique. *J. Artif. Intell. Res.* 16, 321–357 (2002). <https://doi.org/10.1613/jair.953>
38. Grolman, E., et al.: Transfer learning for user action identification in mobile apps via encrypted trafic analysis. *IEEE Intell. Syst.* (2), 40–53 (2018). <https://doi.org/10.1109/mis.2018.111145120>
39. Casquilho, J.P.: On the weighted Gini–Simpson index: estimating feasible weights using the optimal point and discussing a link with possibility theory. *Soft Comput.* (22), 1–8 (2020). <https://doi.org/10.1007/s00500-020-05011-6>
40. University, C.: The Stratosphere IPS Project Dataset [EB]. <https://stratosphereips.org/category/dataset.html>, 2016/2021-05-23
41. Simonyan, K., Zisserman, A.: Very Deep Convolutional Networks for Large-Scale Image Recognition. *Computer Science* (2014)
42. Miller, S., Curran, K., Lunney, T.: Multilayer perceptron neural network for detection of encrypted VPN network traffic. In: International Conference on Cyber Situational Awareness, Data Analytics and Assessment (Cyber SA), pp. 1–8. IEEE (2018)
43. Soleymanpour, S., Sadr, H., Beheshti, H.: An efficient deep learning method for encrypted traffic classification on the web. In: 2020 6th International Conference on Web Research (ICWR), pp. 209–216. IEEE (2020)

How to cite this article: Tang, Z., et al.: Markov-GAN: Markov image enhancement method for malicious encrypted traffic classification. *IET Inf. Secur.* 16(6), 442–458 (2022). <https://doi.org/10.1049/ise2.12071>